Pizzario

Sales Report

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Introduction

Pizzario is a SQL-based pizza sales analysis project that focuses on extracting business insights from sales data. Using SQL concepts like Joins, Aggregate Functions, Subqueries, and Window Functions, the project analyzes customer orders, revenue, pizza categories, and sales trends.

The project is divided into Basic, Intermediate, and Advanced levels to study order patterns, top-selling pizzas, revenue contribution, and overall business performance. It demonstrates how SQL can be used for data analysis and business intelligence in the food industry.

Retrieve the total number of orders placed.

Input :

```
select count(order_id) from orders;
```

Output :

	count(order_id)
▶	2454

Calculate the total revenue generated from pizza sales.

Input :

```sql
SELECT
    round(sum(orders_details.quantity * pizzas.price))
FROM
    orders_details
        JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id;
```

Output :

round(sum(orders_details.quantity * pizzas.price))
21493

Identify the highest-priced pizza.

Input :

```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
WHERE
    price = (SELECT
            MAX(price)
        FROM
            pizzas);
```

Output :

name	price
The Greek Pizza	35.95

Identify the most common pizza size ordered.

Input :

```
SELECT
    pizzas.size, COUNT(orders_details.order_details_id) AS count
FROM
    pizzas
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY size
ORDER BY count DESC
LIMIT 1;
```

Output :

	size	count
▶	L	504

List the top 5 most ordered pizza types along with their quantities.

Input :

```
SELECT
    pizza_types.name AS name,
    COUNT(orders_details.quantity) AS count
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY name
ORDER BY count DESC
LIMIT 5;
```

Output :

name	count
The Pepperoni Pizza	70
The Thai Chicken Pizza	66
The Classic Deluxe Pizza	64
The Hawaiian Pizza	62
The Italian Supreme Pizza	60

Join the necessary tables to find the total quantity of each pizza category ordered.

Input :

```
SELECT
    pizza_types.category AS category,
    sum(orders_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY category
ORDER BY quantity ASC;
```

Output :

category	quantity
Chicken	280
Veggie	312
Supreme	315
Classic	392

Determine the distribution of orders by hour of the day

Input :

```
SELECT
    HOUR(order_time) AS hour, COUNT(order_id)
FROM
    orders
GROUP BY hour;
```

Output :

Result Grid | Filter Ro

hour	COUNT(order_id)
11	137
12	278
13	266
14	216
15	172
16	216
17	271
18	275
19	226
20	193
21	133
22	70

Join relevant tables to find the category-wise distribution of pizzas.

Input :

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Output :

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Group the orders by date and calculate the average number of pizzas ordered per day.

Input :

```
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date AS date,
            SUM(orders_details.quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY date) AS order_quantity;
```

Output :

	ROUND(AVG(quantity), 0)
▶	144

Determine the top 3 most ordered pizza types based on revenue.

Input :

```
SELECT
    pizza_types.name AS name,
    SUM(orders_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY name
ORDER BY revenue DESC
LIMIT 3;
```

Output :

name	revenue
The Thai Chicken Pizza	1229.5
The Italian Supreme Pizza	1071.75
The Barbecue Chicken Pizza	1056.25

Calculate the percentage contribution of each pizza type to total revenue.

Input :

```
SELECT
    pizza_types.category AS category,
    ROUND(SUM(orders_details.quantity * pizzas.price) / (SELECT
                    ROUND(SUM(orders_details.quantity * pizzas.price),
                                2)
                FROM
                    orders_details
                        JOIN
                    pizzas ON orders_details.pizza_id = pizzas.pizza_id) * 100,
            2) AS revenue_percentage
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY category
ORDER BY revenue_percentage DESC;
```

Output :

category	revenue_percentage
Classic	26.63
Supreme	25.56
Veggie	24.47
Chicken	23.33

Analyze the cumulative revenue generated over time.

Input :

```
select order_date, sum(revenue) over (order by order_date) as cum_revenue from (SELECT
    orders.order_date,
    SUM(orders_details.quantity * pizzas.price) AS revenue
FROM
    orders_details
        JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
        JOIN
    orders ON orders.order_id = orders_details.order_id
GROUP BY orders.order_date) as sales;
```

Output :

Result Grid | Filter Rows:

order_date	cum_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21493.15